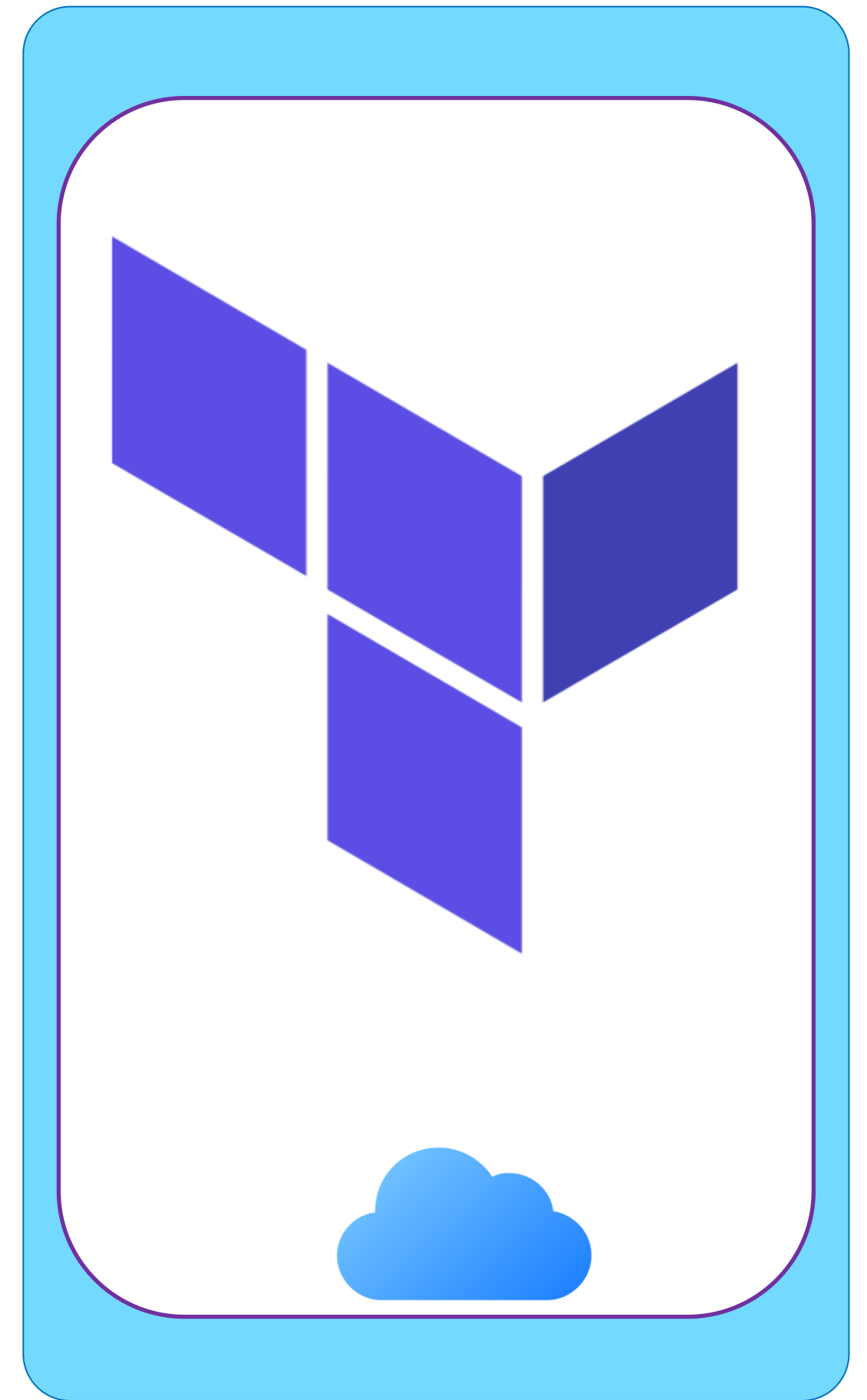


Mastering Terraform Cloud I

A Hands-on Beginners Guide



Mastering Terraform Cloud | A Hands-on Beginners Guide

- ❖ Learn Infrastructure as Code with Terraform Cloud
- ❖ Understand Terraform Open Source, Cloud & Enterprise
- ❖ Explore hands-on Terraform Cloud demonstrations
- ❖ Work with:
 - Workspaces & Projects
 - Variables
 - Remote Runs
 - Cost Estimation
 - Notifications
 - Access Management



**The Ultimate Linux Bootcamp for
DevOps SRE & Cloud Engineers**



**Practical Kubernetes –
Beyond CKA and CKAD**



**Automation with Ansible -
Hands-on DevOps**



**Kubernetes and
Cloud Native Associate**



**Argo CD for the Absolute
Beginners - Hands-On**



**Mastering Docker Essentials
- Hands-on**



Kulbhushan Mayer



Yogesh Raheja

How exactly does this course work?



Learn concepts and apply them through hands-on demonstrations

Create and configure Terraform Cloud projects, workspaces & AWS credentials

Connect Terraform Cloud with configurations and automate plan & apply changes

Course Workflow

IaC & Terraform — Terraform offerings: Open Source, Cloud & Enterprise

Terraform Cloud — Organizations, Projects & Workspaces

Hands-On Setup — Workspaces, AWS credentials & Terraform plans

Automation — Configurations & automatic infrastructure changes

Advanced Features — Speculative plans, PR workflows & cost estimation

Variables & Notifications — HCL variables & Microsoft Teams notifications

Access & Resource Management — Access control, resource & workspace management

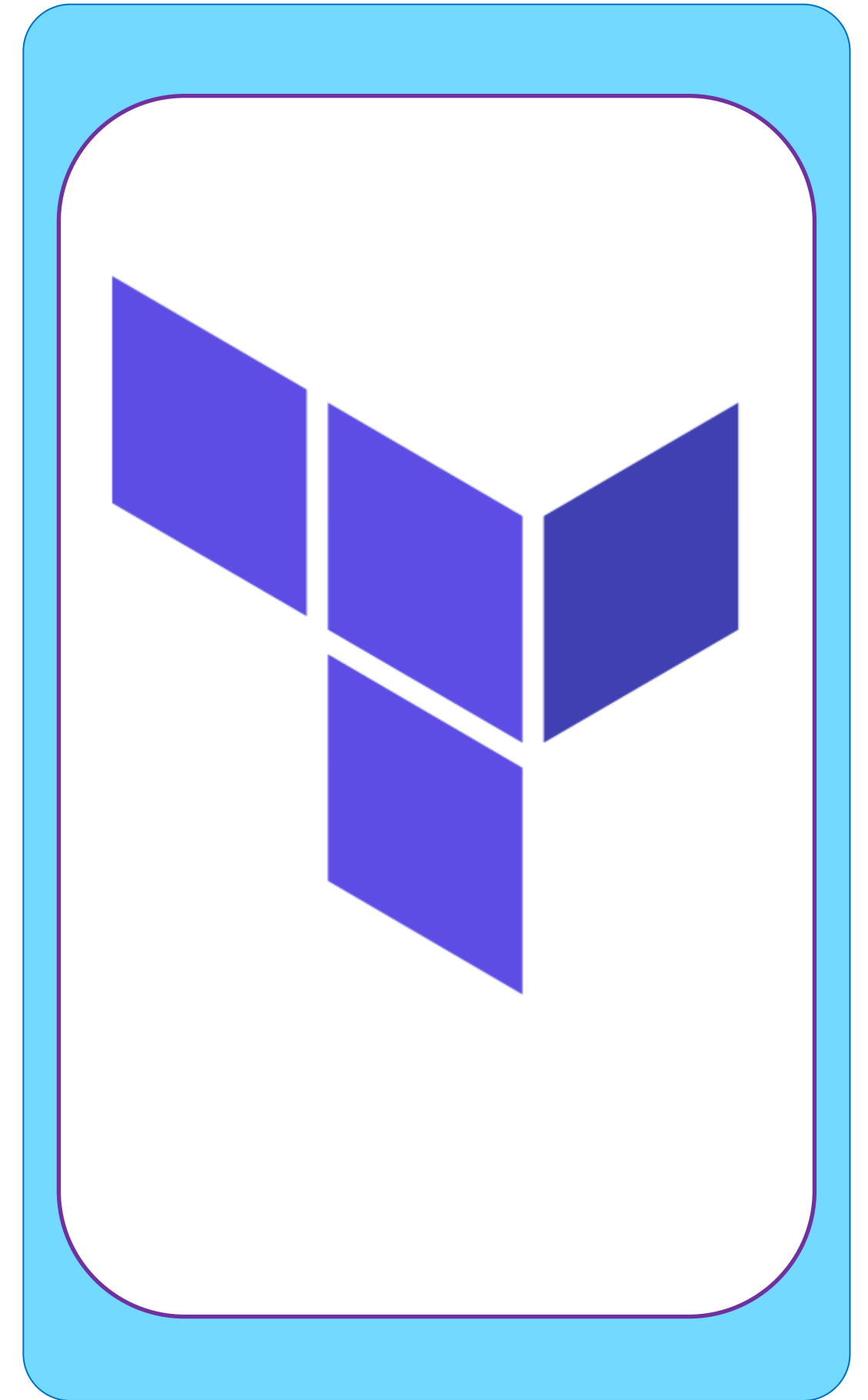
Goal & Prerequisites — Practical IaC foundation; basic Terraform, cloud & AWS knowledge



THANK YOU

FOR ATTENTION

Understanding IaC & Terraform Cloud



Understanding IaC & Terraform Cloud

❖ Section Introduction

- Understand Infrastructure as Code
- Explore Terraform & its Editions
- Learn Organizations, Projects & Workspaces
- Understand Free Tier Capabilities & Limitations
- Explore Real-World Use Cases
- Build Terraform Fundamentals



What is Infrastructure as Code (IaC)

What is Infrastructure as Code (IaC)



Traditional Approach

- Manual infrastructure setup
- Repeated console/server steps
- Time-consuming and error-prone
- Difficult to reproduce consistently

What is Infrastructure as Code (IaC)



Modern DevOps

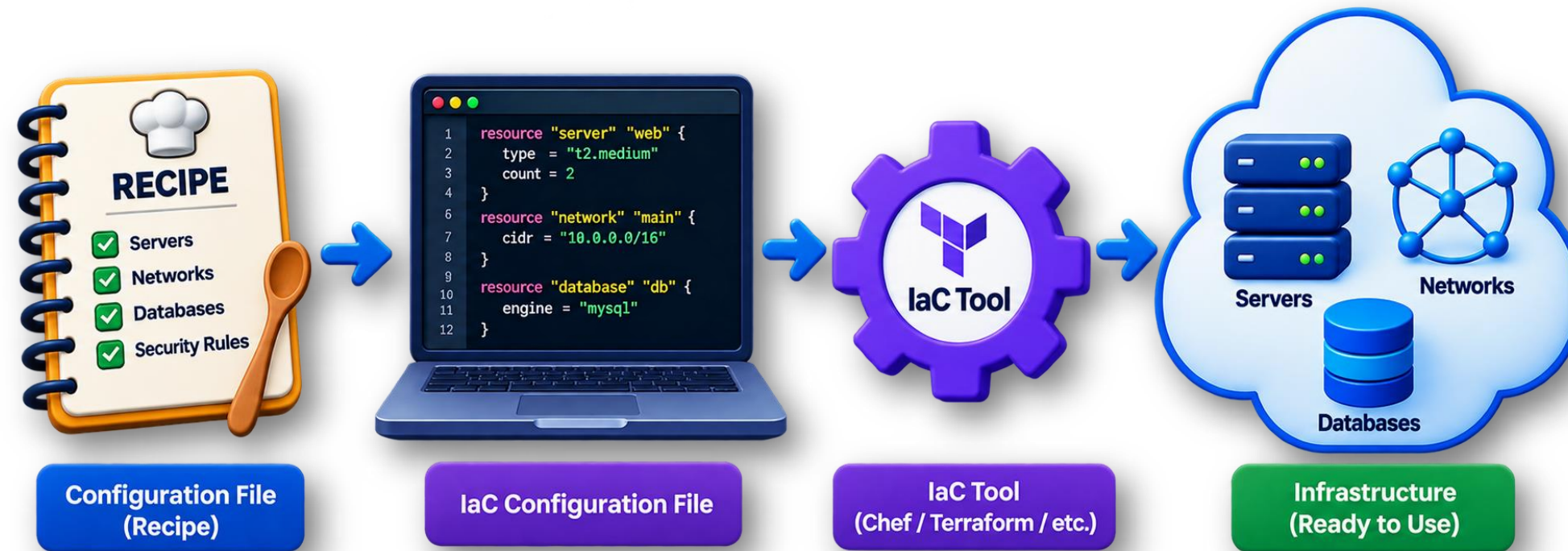
- Manage infrastructure like software
- Write code instead of manual console actions



Infrastructure as Code (IaC)

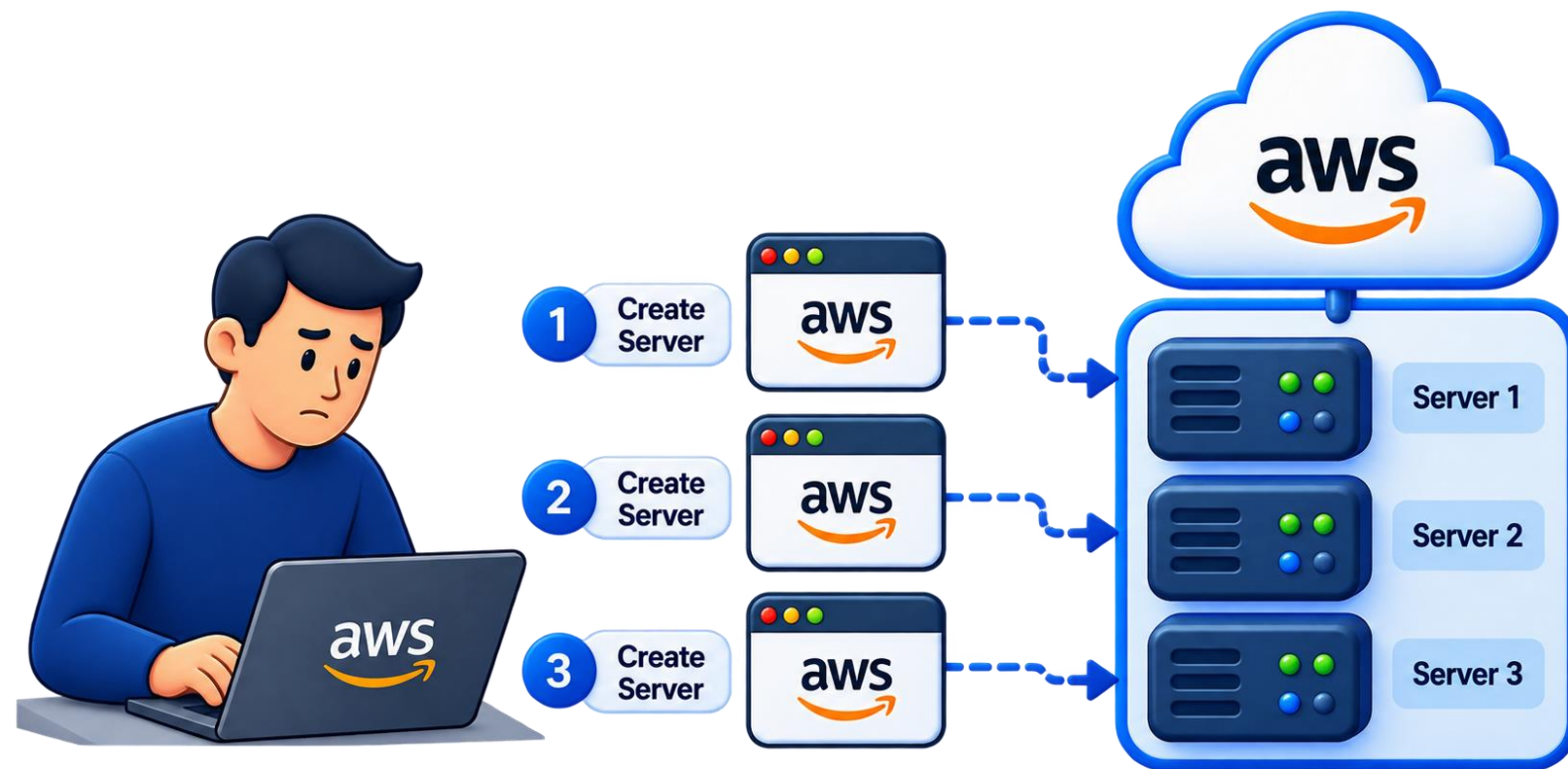
- Define infrastructure using configuration files
- Provision infrastructure without manual steps

What is Infrastructure as Code (IaC)



What is Infrastructure as Code (IaC)

Manual Approach



IaC Approach with Terraform



Consistency

Reduces Human Error

Faster Deployments

What is Infrastructure as Code (IaC)

Summary

- IaC turns infrastructure management into software development
- Provision resources quickly and reliably



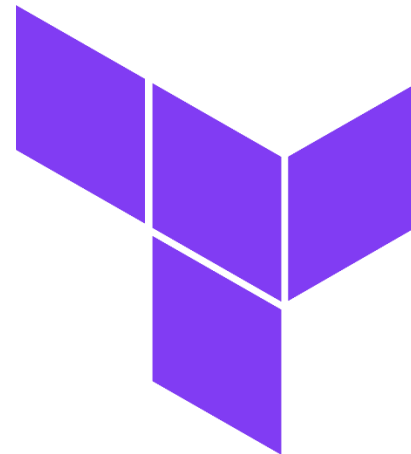
What is Terraform

What is Terraform

Infrastructure as Code, or IaC



Which tool can help us implement Infrastructure as Code in real-world environments?



What is Terraform

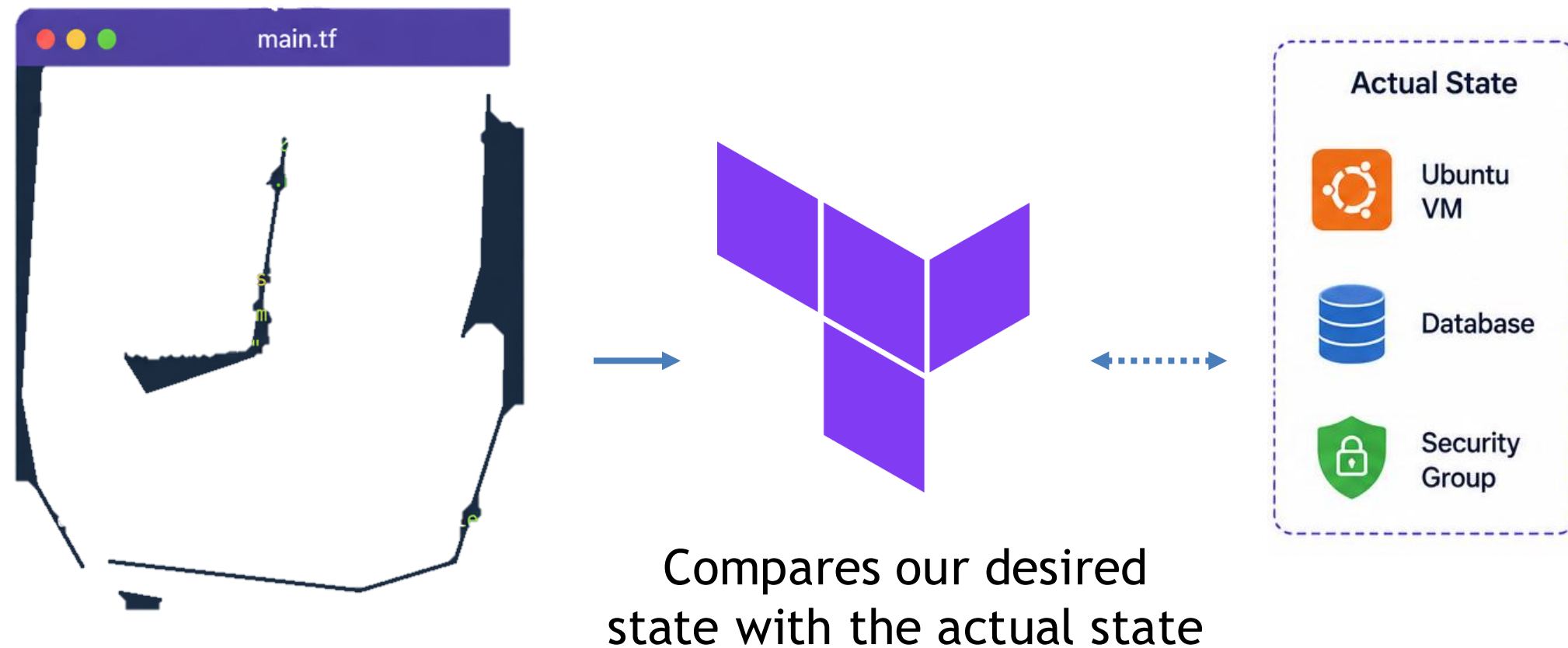


- Infrastructure as Code tool
- Developed by HashiCorp
- Build, modify & manage infrastructure
- Uses simple configuration files
- Uses HashiCorp Configuration Language (HCL)
- Human-readable & easy to understand

What is Terraform

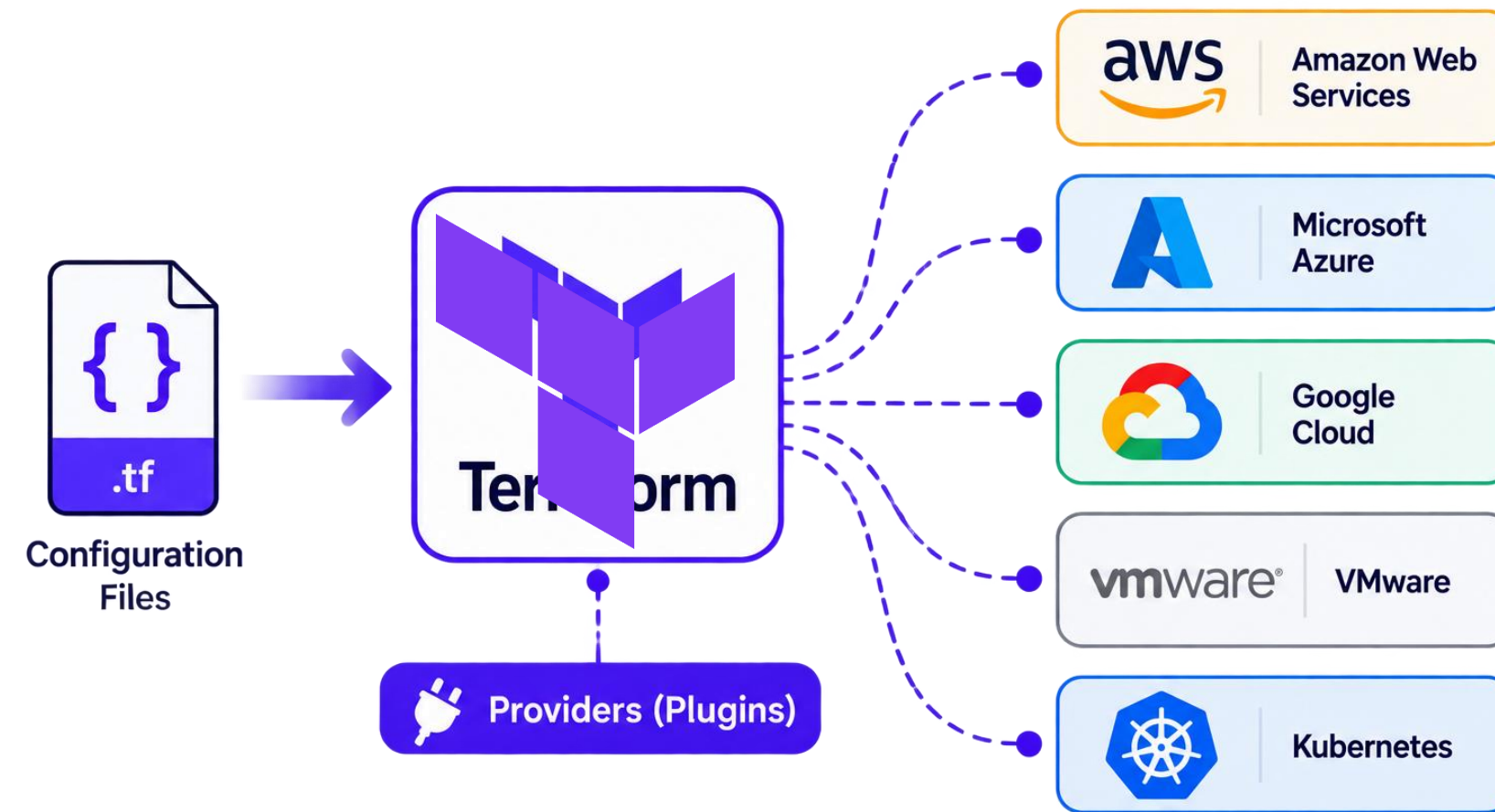
Example

- One Ubuntu virtual machine
- One database
- One security group



Declare what you want – Terraform figures out how to create it

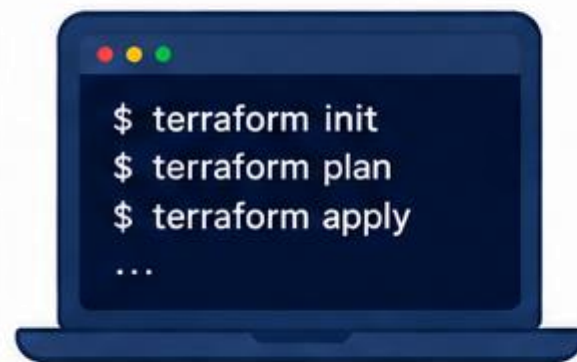
What is Terraform



What is Terraform

Terraform was first released in 2014

1. Terraform Open Source (Terraform CLI)



 Free to use

2. Terraform Cloud (HCP Terraform)



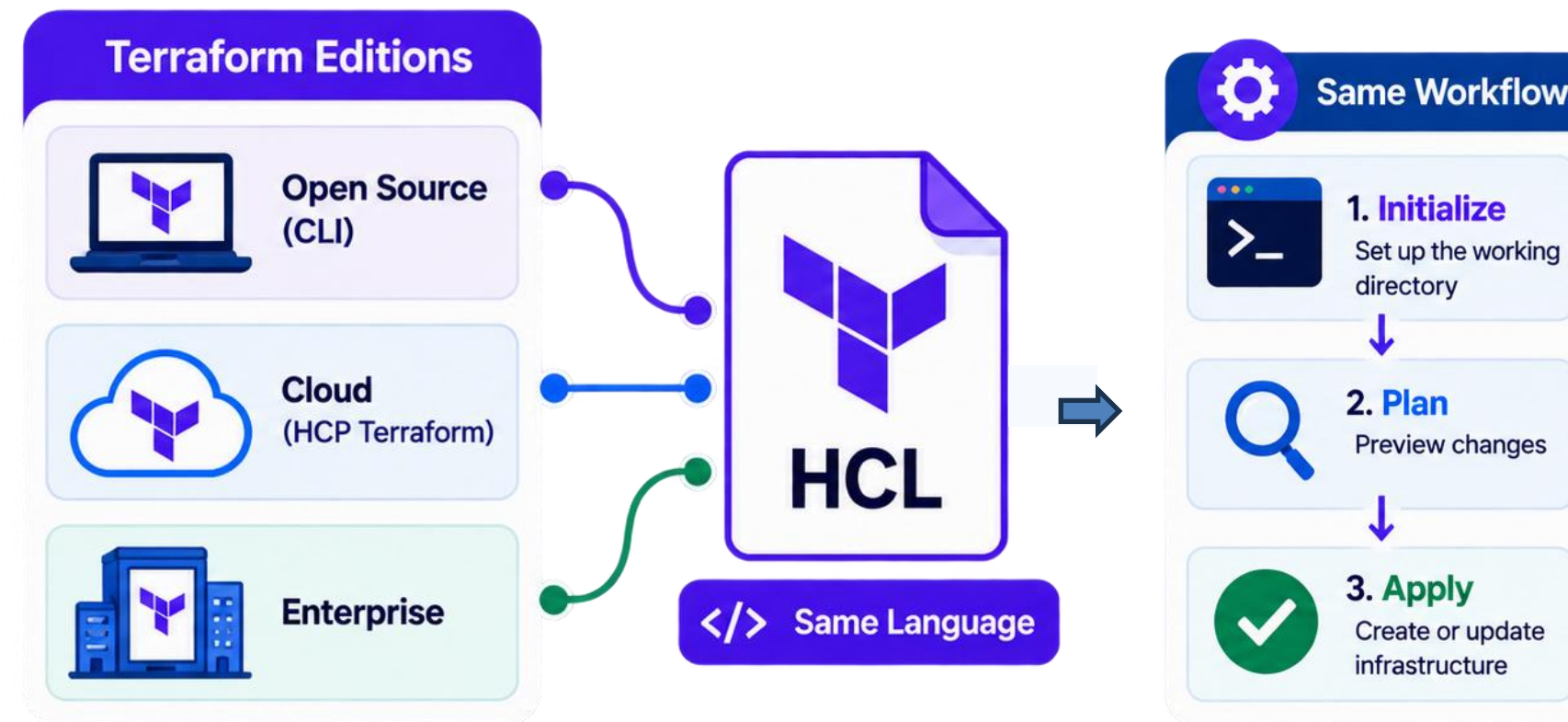
 Fully managed by HashiCorp

3. Terraform Enterprise



 Self-hosted

What is Terraform



What is Terraform

Summary

- Powerful Infrastructure as Code tool
- Define infrastructure in simple, readable files
- Automate provisioning across cloud platforms



Difference between Terraform OSS, Cloud, Enterprise

Terraform OSS, Cloud, Enterprise



What exactly is the difference between them?



Note

Terraform Cloud is now part of HashiCorp Cloud Platform, also known as HCP

Terraform Cloud is often referred to as HCP Terraform

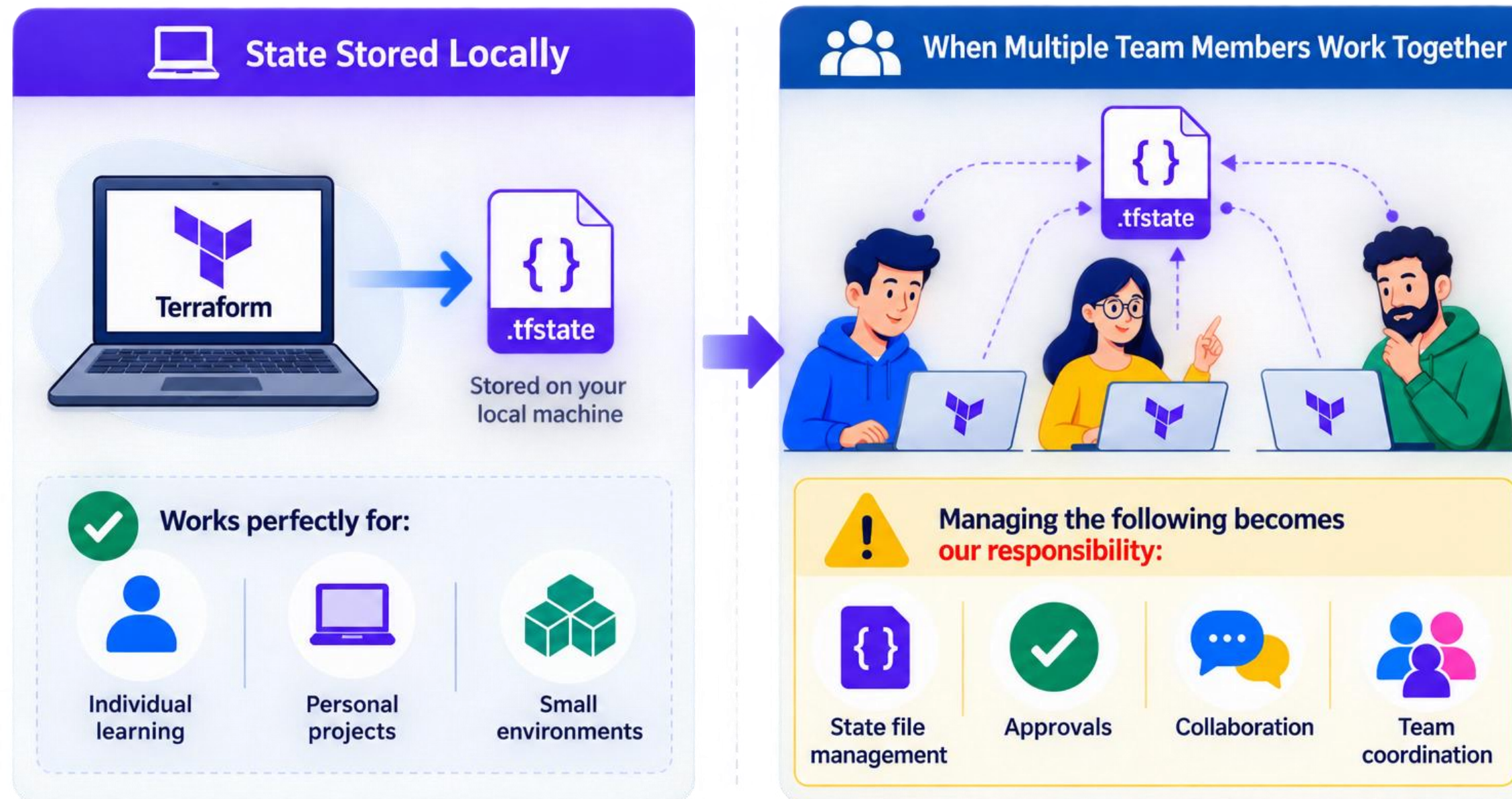
Terraform OSS, Cloud, Enterprise

❑ Terraform Open Source

- Free version of Terraform
- Runs on your local machine
- Includes core commands:
 - **terraform init**
 - **terraform plan**
 - **terraform apply**
- Provision infrastructure across multiple cloud providers
- Uses the same HCL configuration language

Terraform OSS, Cloud, Enterprise

❑ Terraform Open Source



- Store the state file in a shared location
- Use Git and CI/CD pipelines to coordinate changes

Terraform OSS, Cloud, Enterprise

❑ Terraform Cloud

Terraform Cloud is a fully managed service hosted by HashiCorp



It builds on top of **Terraform Open Source** and adds powerful collaboration and automation features

❖ Features include:

Remote state storage

Git integration

Team-based
access controls

Private module
registry

Web-based user
interface

Run history and logs

Terraform OSS, Cloud, Enterprise

❑ Terraform Cloud

- Push code to Git repository
- Terraform Cloud automatically runs Plan & Apply
- Enables safe team collaboration
- Prevents overwriting each other's work

No servers or infrastructure to maintain

HashiCorp manages everything

Terraform OSS, Cloud, Enterprise

❑ Terraform Enterprise

- Same core capabilities as Terraform Cloud
- Installed in own data center or private cloud
- Designed for large organizations
- Supports security, compliance & data residency requirements

❖ Advanced Capabilities:

Role-Based Access Control, or RBAC

Single Sign-On using SAML or LDAP

Audit logging

Enhanced security and compliance controls

No resource or concurrency limits

Terraform OSS, Cloud, Enterprise

Summary

- **Terraform Open Source** — Free CLI for individuals & small projects
- **Terraform Cloud (HCP Terraform)** — Managed SaaS with collaboration, automation & governance
- **Terraform Enterprise** — Self-hosted with advanced enterprise controls

Terraform OSS, Cloud, Enterprise

Terraform OSS — Free and runs locally

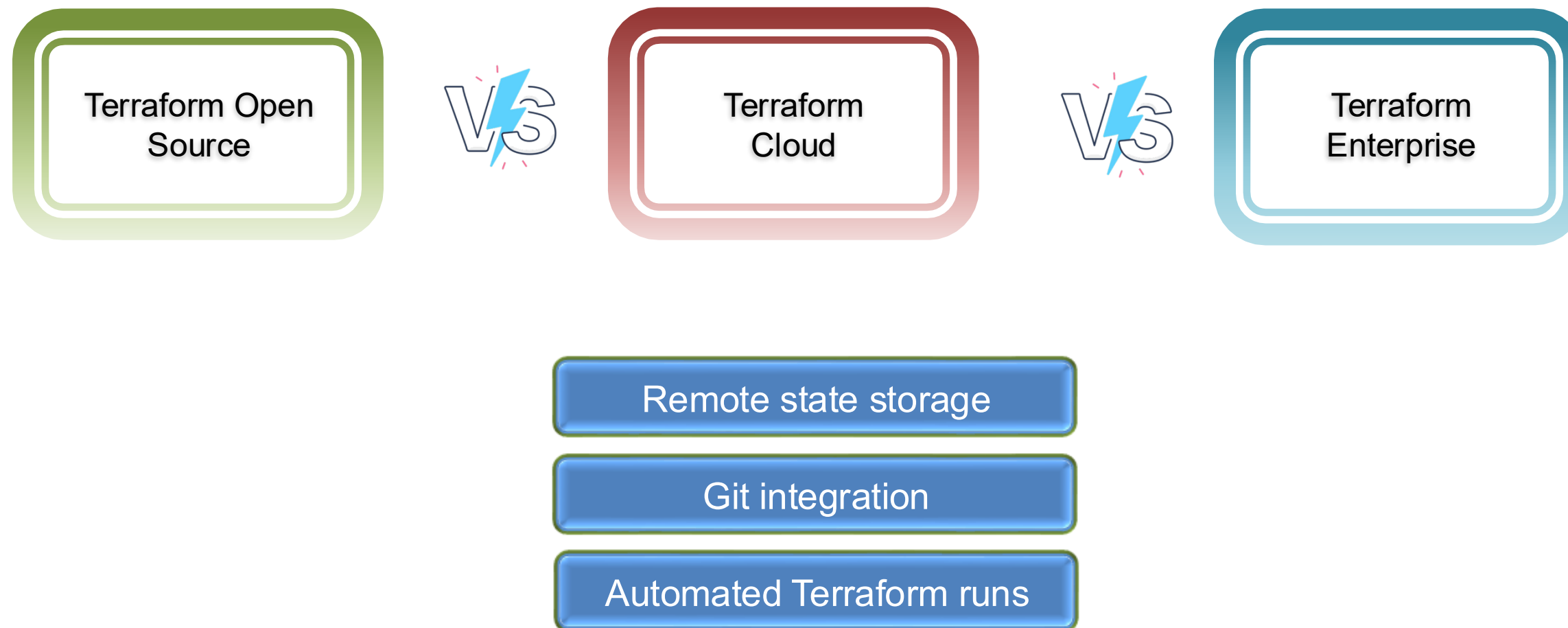
Terraform Cloud / HCP Terraform — Managed by HashiCorp

Terraform Enterprise — Managed by our organization



Terraform Cloud overview (Organizations, Projects & Workspaces)

Terraform Cloud overview (Organizations, Projects & Workspaces)



Terraform Cloud overview (Organizations, Projects & Workspaces)



How does Terraform Cloud organize all of our infrastructure and environments?

- Terraform Cloud Hierarchy
 - Organizations
 - Projects
 - Workspaces

Terraform Cloud overview (Organizations, Projects & Workspaces)

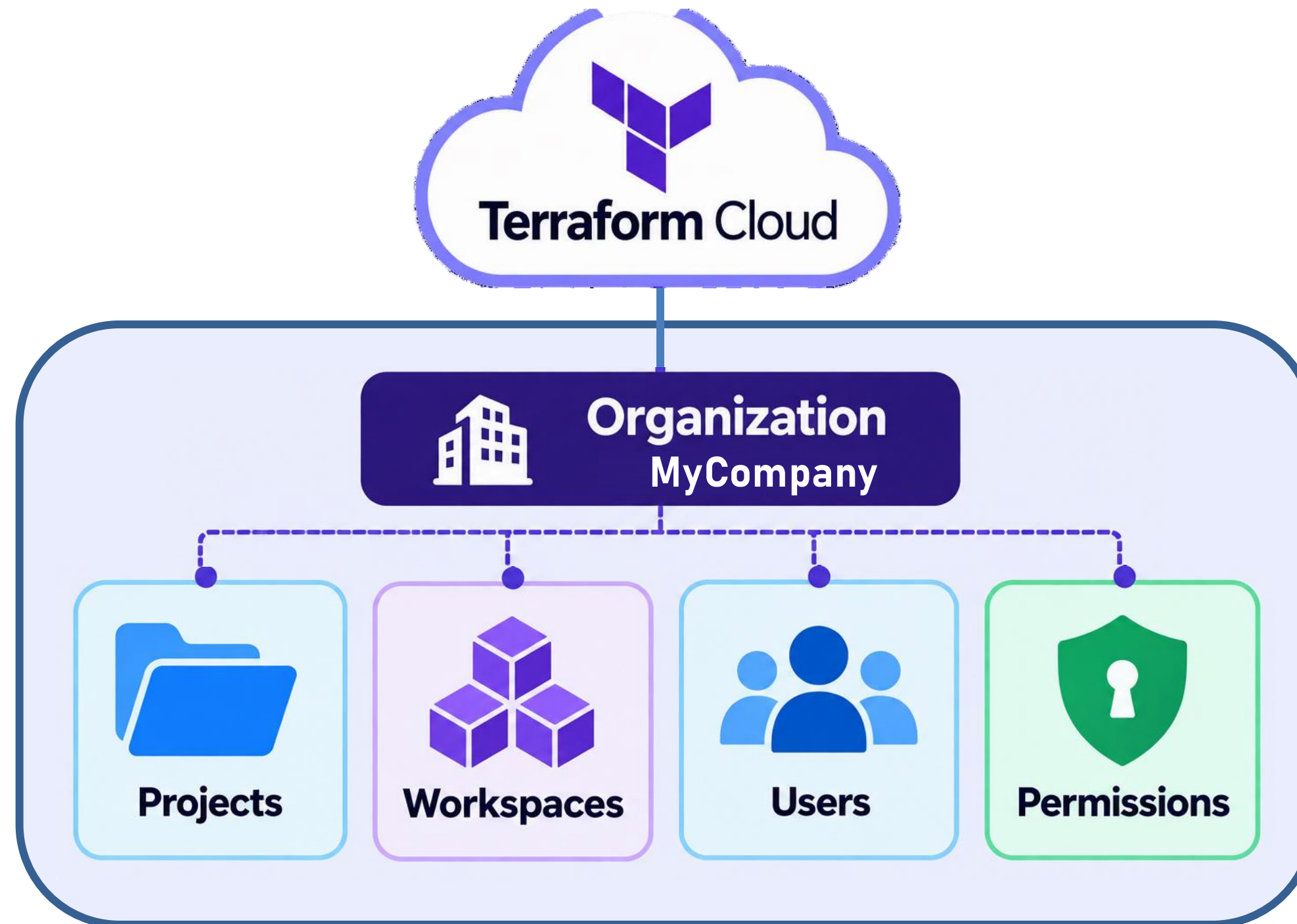
❑ Organization

An **Organization** is the highest-level container in Terraform Cloud



Terraform Cloud overview (Organizations, Projects & Workspaces)

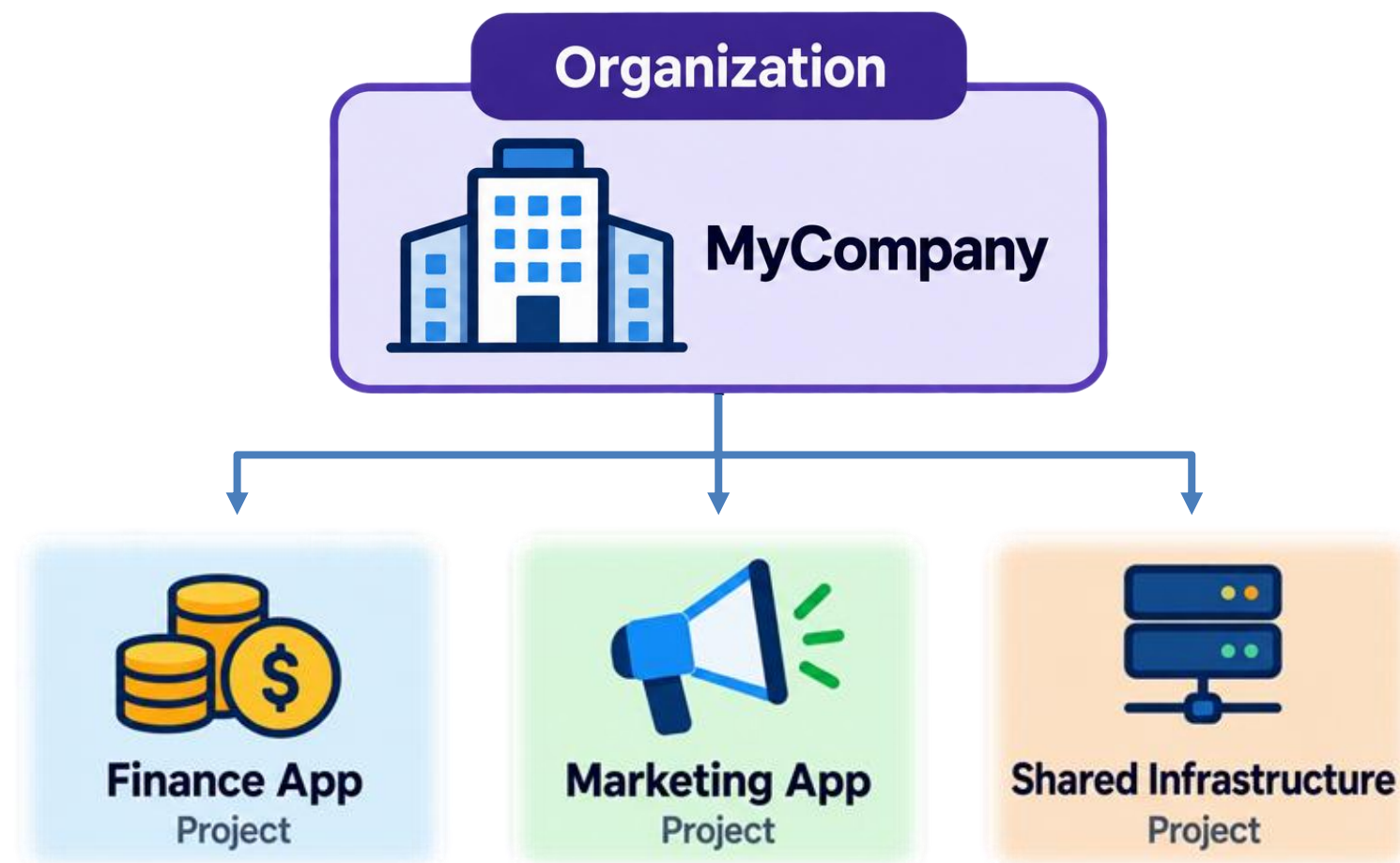
❑ Organization



Terraform Cloud overview (Organizations, Projects & Workspaces)

❑ Projects

Projects help us group related workspaces together



- Organize infrastructure across multiple workspaces
- Apply common settings and permissions
- Separate development and production environments

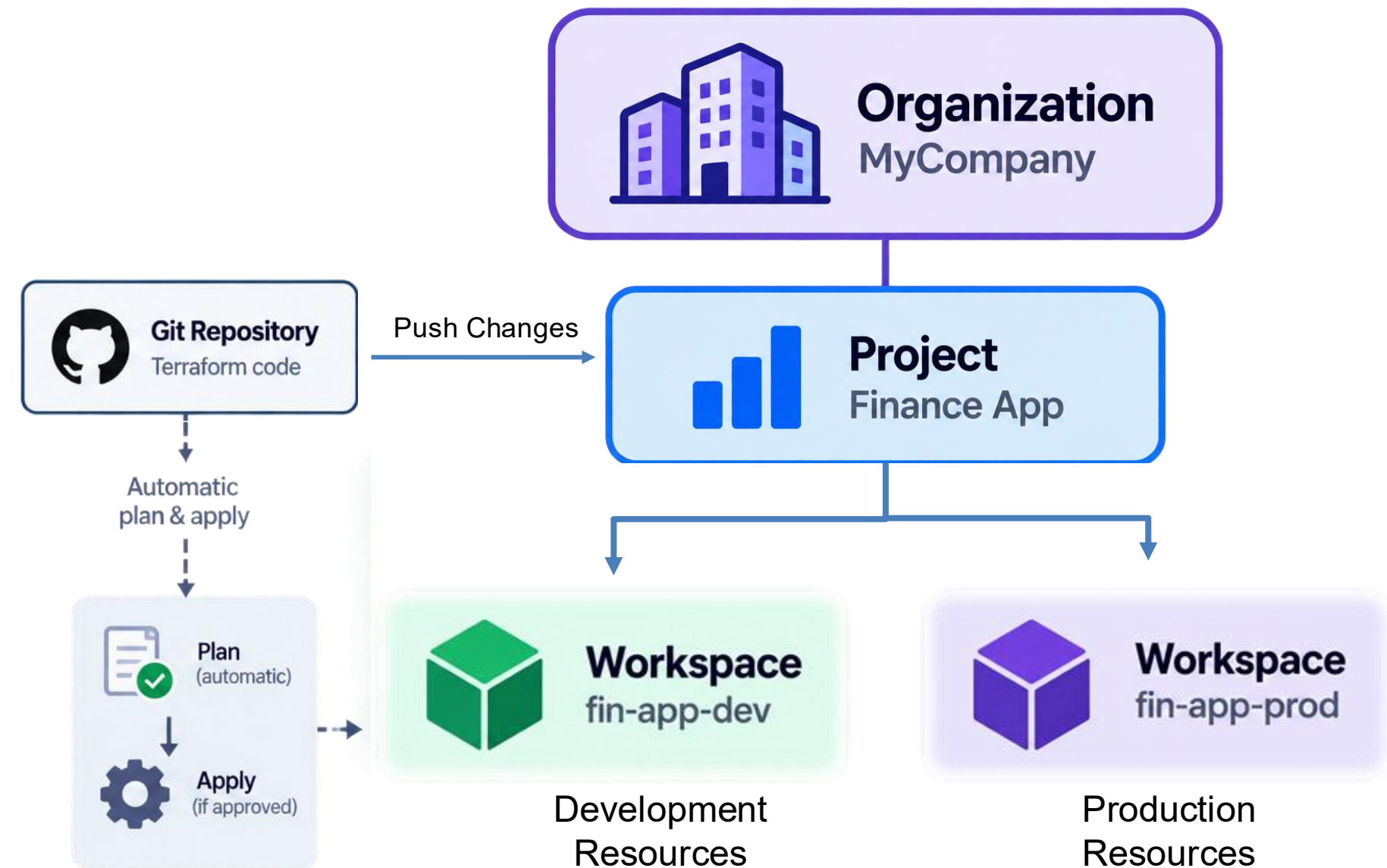
Terraform Cloud overview (Organizations, Projects & Workspaces)

❑ Workspaces

▪ Each workspace contains:

- Our Terraform configuration
- Its own state file
- Run history
- Variables and settings

A workspace represents one specific piece of infrastructure



- Each workspace maintains its own state file
- Enables safe team collaboration
- Prevents overwriting each other's changes

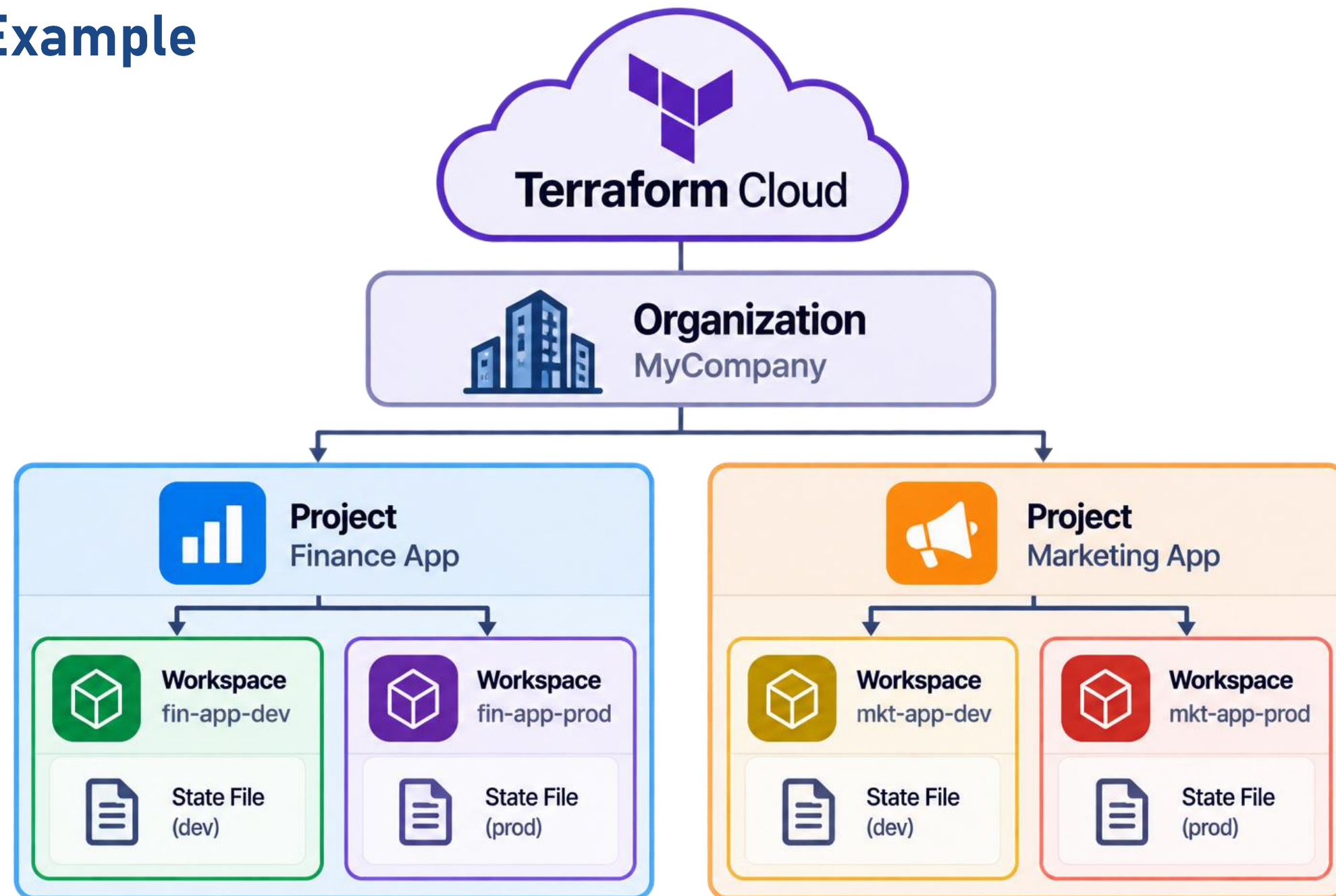
Terraform Cloud overview (Organizations, Projects & Workspaces)

Organization → Projects → Workspaces

- **Organization** = Our company or team
- **Project** = A group of related applications or environments
- **Workspace** = The place where Terraform runs and manages infrastructure

Terraform Cloud overview (Organizations, Projects & Workspaces)

Example



Organized



Scalable



Easy to Manage

Terraform Cloud overview (Organizations, Projects & Workspaces)

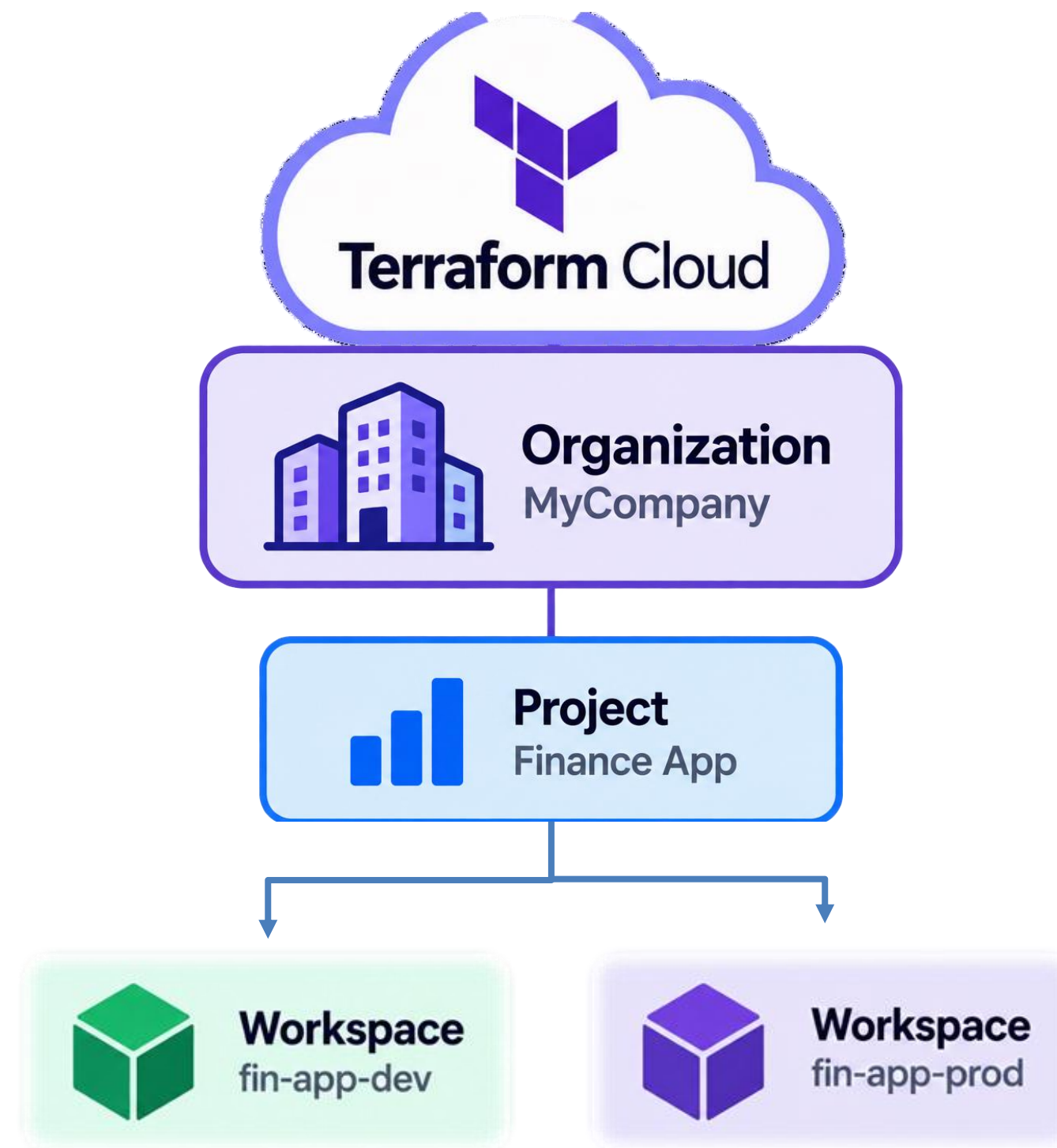
Summary

- **Organizations** — Define the top-level boundary
- **Projects** — Group related workspaces
- **Workspaces** — Execute Terraform and manage infrastructure



Free Tier capabilities and limitations

Free Tier capabilities and limitations



Free Tier capabilities and limitations

❑ Capabilities

Free tier includes many powerful features that are more than enough for learning and small team projects

Store Terraform state remotely and securely

Run Terraform plans and applies in the cloud

Connect Git repositories such as GitHub and GitLab

Collaborate with multiple team members

Use a private module registry

View run history and logs in the web interface

Free tier provides most of the core features we need to learn Terraform Cloud and work effectively as a team

Free Tier capabilities and limitations

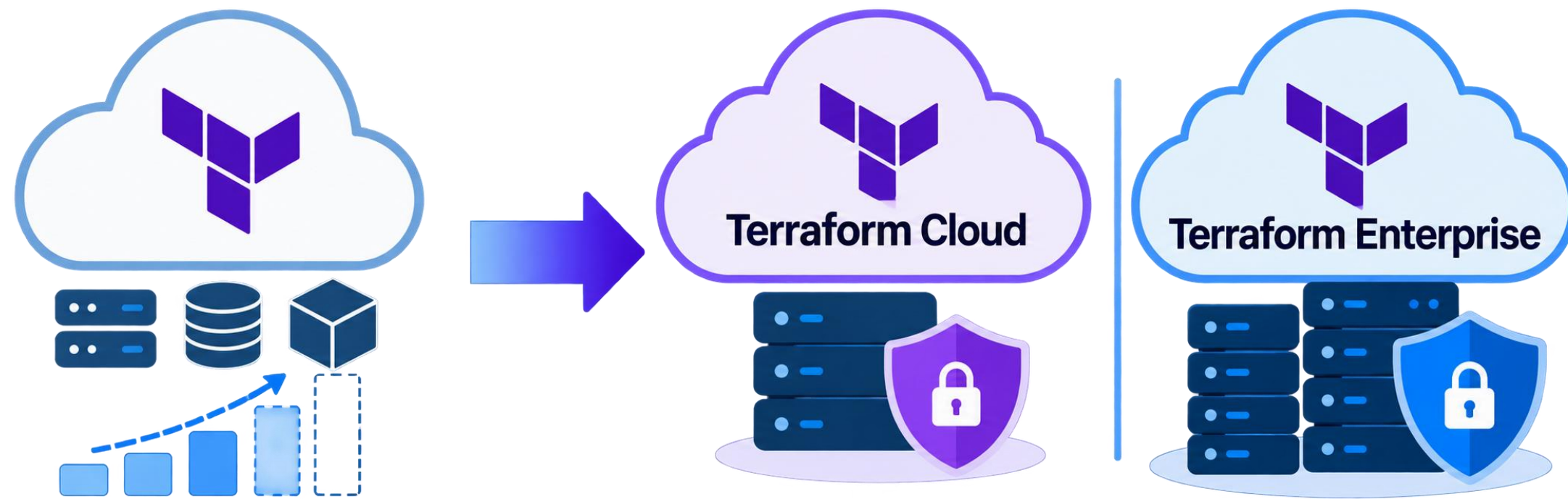
❑ Limitations

- **Managed resources:** Up to 500 resources
- **Concurrent runs:** Only one run at a time
- Additional runs **wait** until the current run completes
- Other limits may apply to:
 - **Usage**
 - **Retained state versions**
 - **Advanced enterprise capabilities**

Limits may change over time

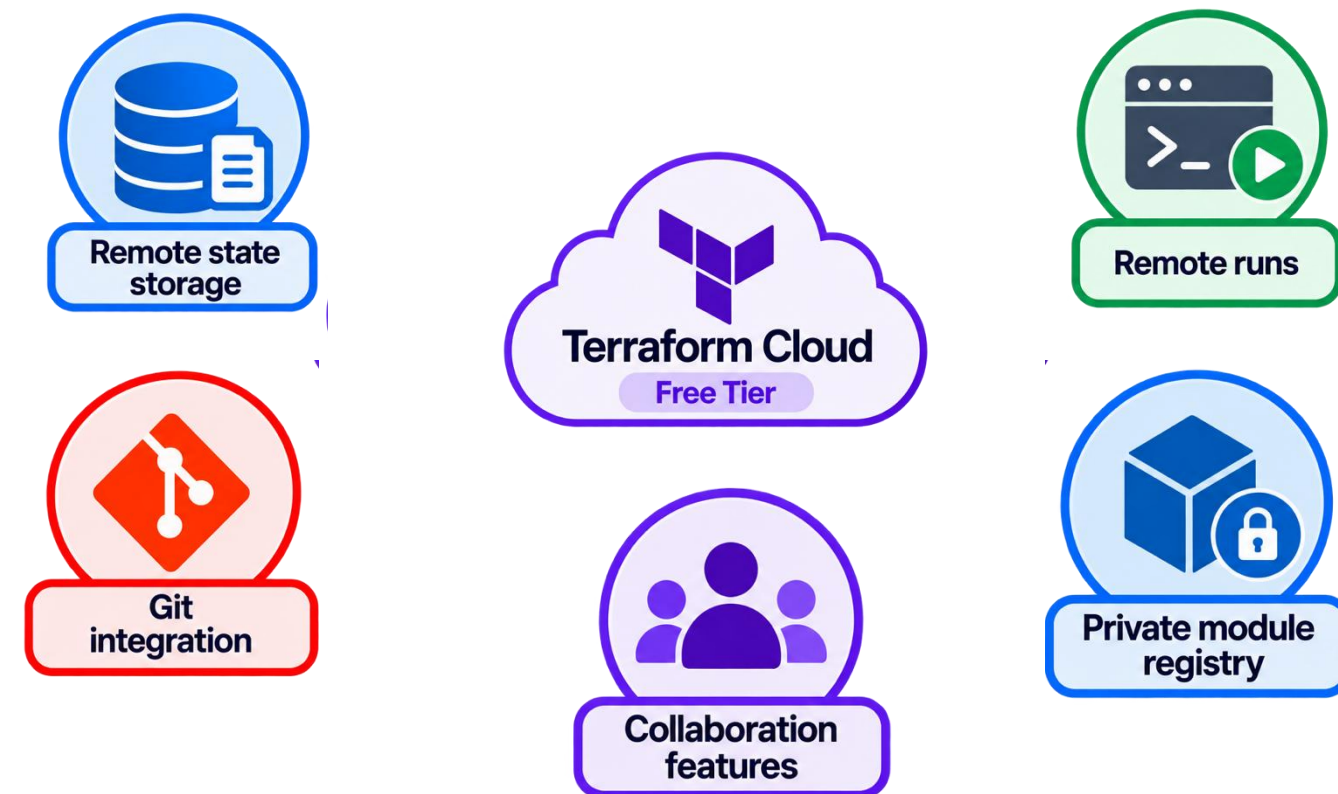
Most important for learners and small projects

Free Tier capabilities and limitations



Free Tier capabilities and limitations

Summary



Example

- Up to 500 managed resources
- One concurrent Terraform run

Free Tier capabilities and limitations

Summary





Real-world use cases

Real-world use cases

☐ Real-World Use Cases of Terraform

- **Multi-Cloud Deployments**
 - Manage AWS, Azure & Google Cloud
 - Same language and workflow
- **Complete Application Stacks**
 - Provision databases, servers, load balancers & storage
 - Automatically handles dependencies
- **Self-Service Infrastructure**
 - Reusable modules
 - Approved environments with less manual work

Real-world use cases

❑ Real-World Use Cases of Terraform

▪ Security & Compliance

- Enforce tagging standards
- Deploy resources in approved regions

▪ Temporary Environments

- Quickly create development, testing & demo environments
- Destroy when no longer needed

▪ Beyond VMs & Networking

- Kubernetes clusters
- Databases
- DNS records
- Platform services

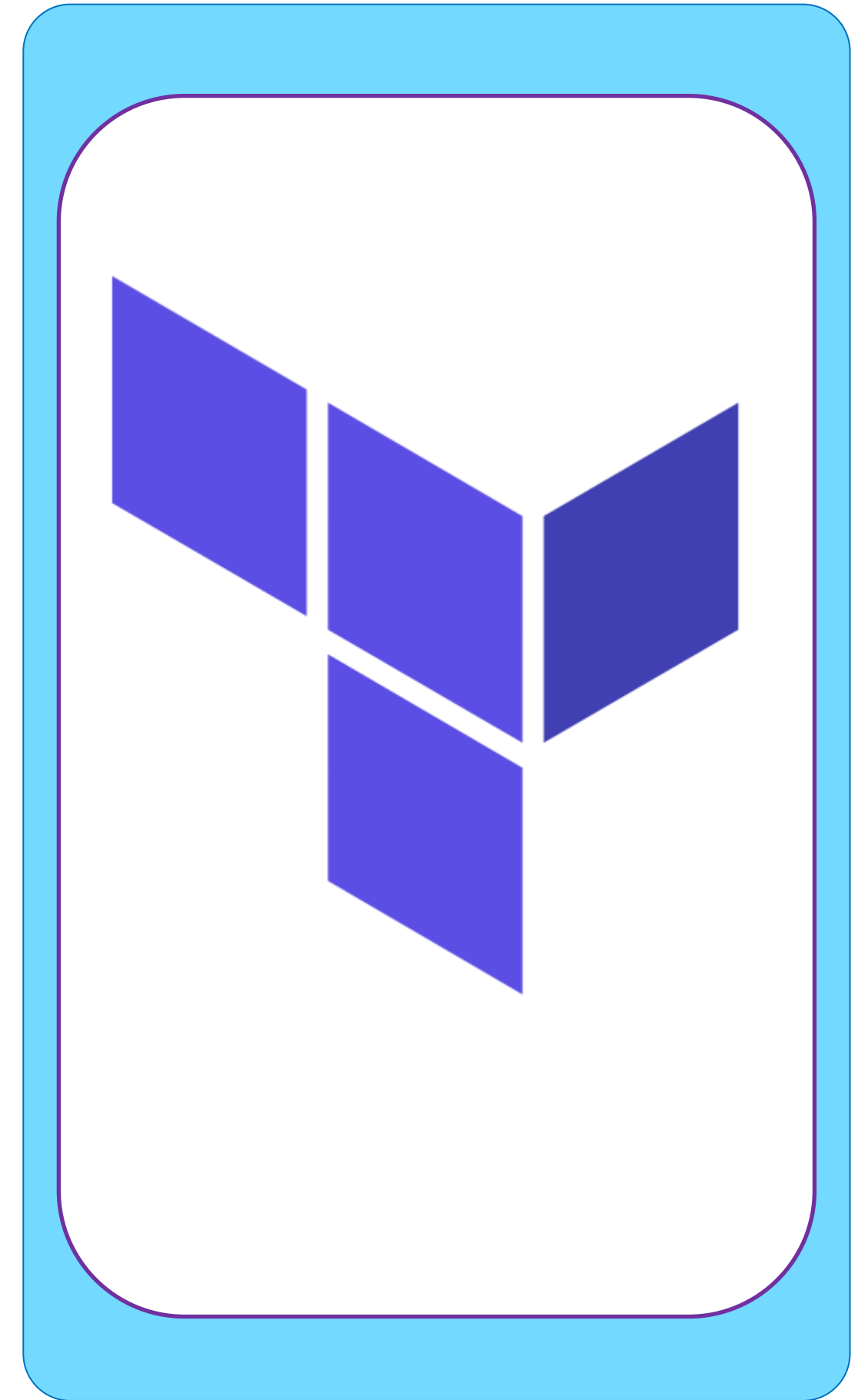
*If a platform provides an API,
Terraform can usually manage it*

Real-world use cases

Summary

- Multi-cloud deployments
- Complete application stacks
- Self-service infrastructure
- Policy and compliance enforcement
- Temporary environments

Understanding IaC & Terraform Cloud



Understanding IaC & Terraform Cloud

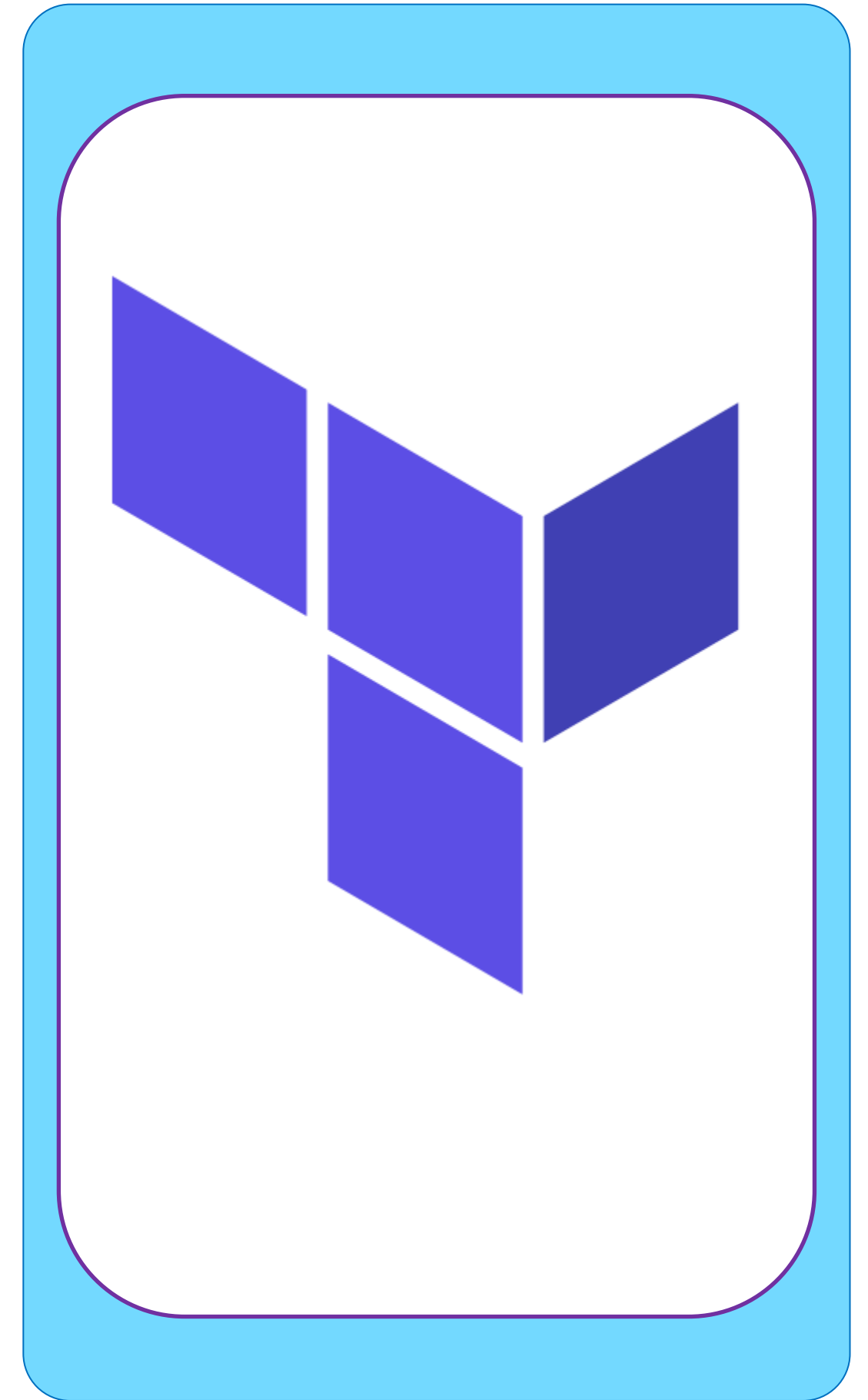
❖ Section Summary

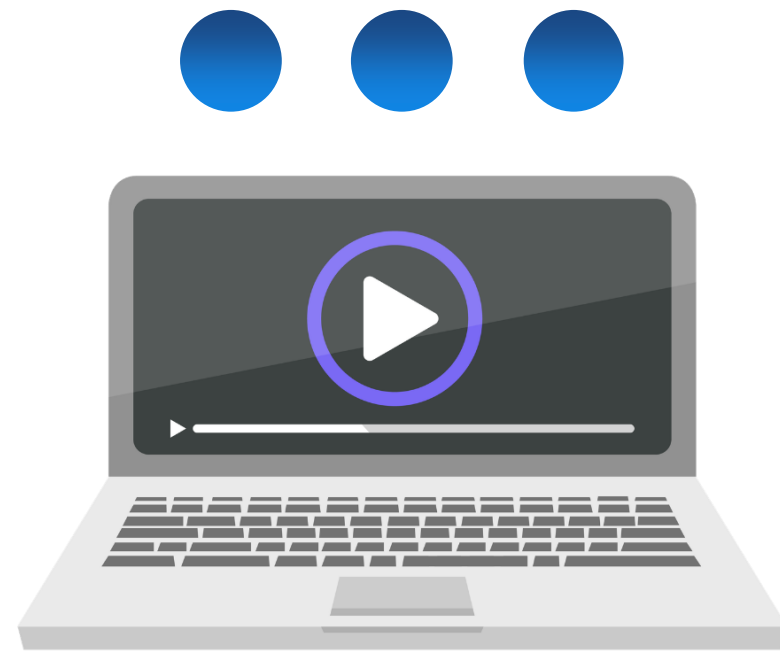
- **Infrastructure as Code** — Manage infrastructure using configuration files
- **Terraform** — Core concepts & different editions
- **Terraform Cloud** — Organizations, Projects & Workspaces
- **Free Tier** — Capabilities & limitations
- **Real-World Use Cases** — Terraform across environments & platforms

THANK YOU

FOR ATTENTION

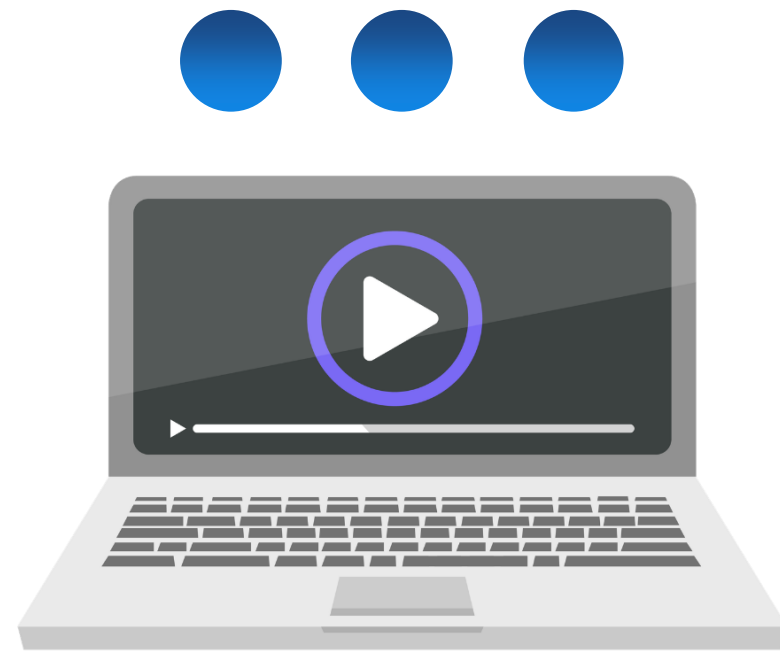
Getting Started with Terraform Cloud





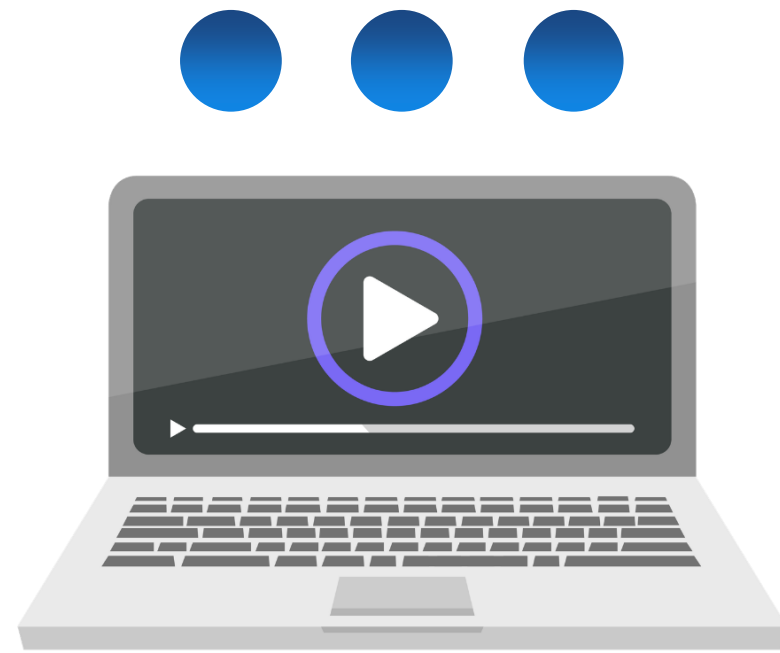
Demo

Terraform Cloud Account Setup



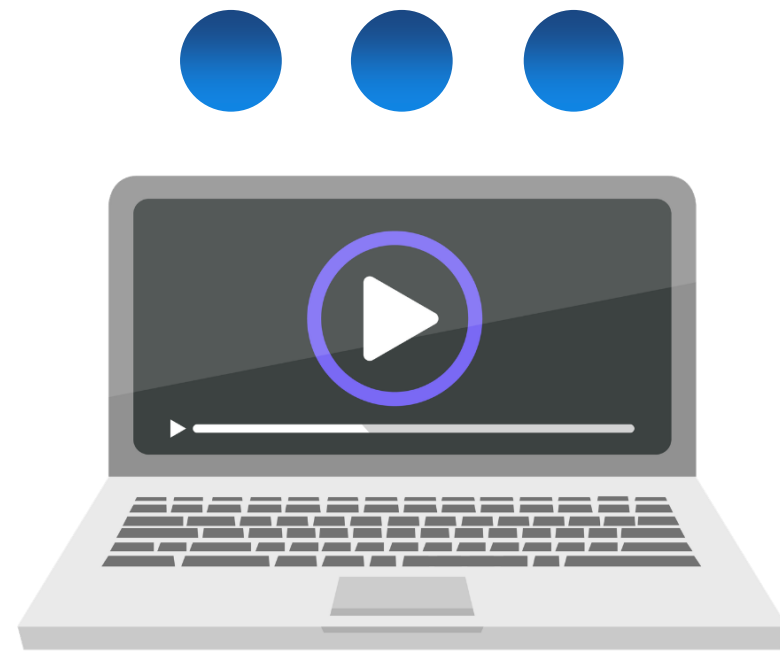
Demo

Exploring Terraform Cloud UI



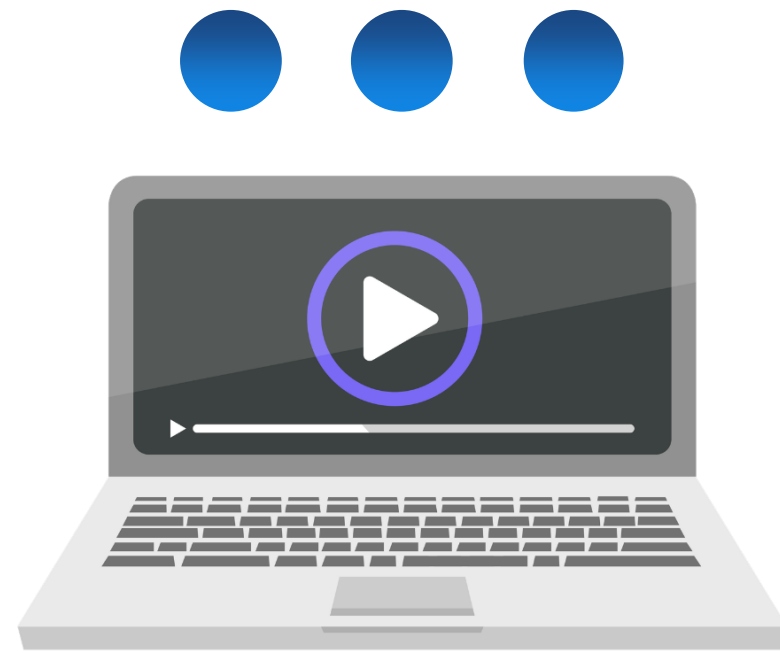
Demo

Creating New Project



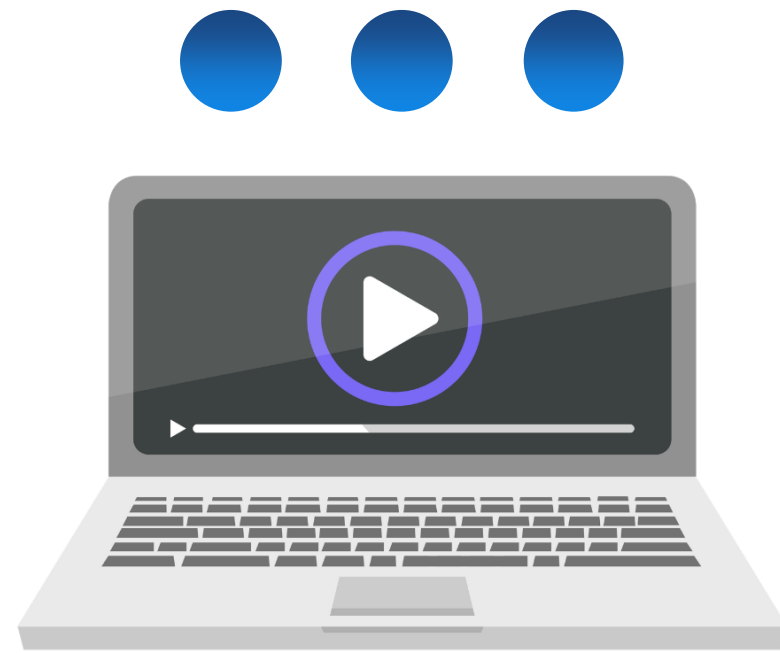
Demo

Creating First Workspace



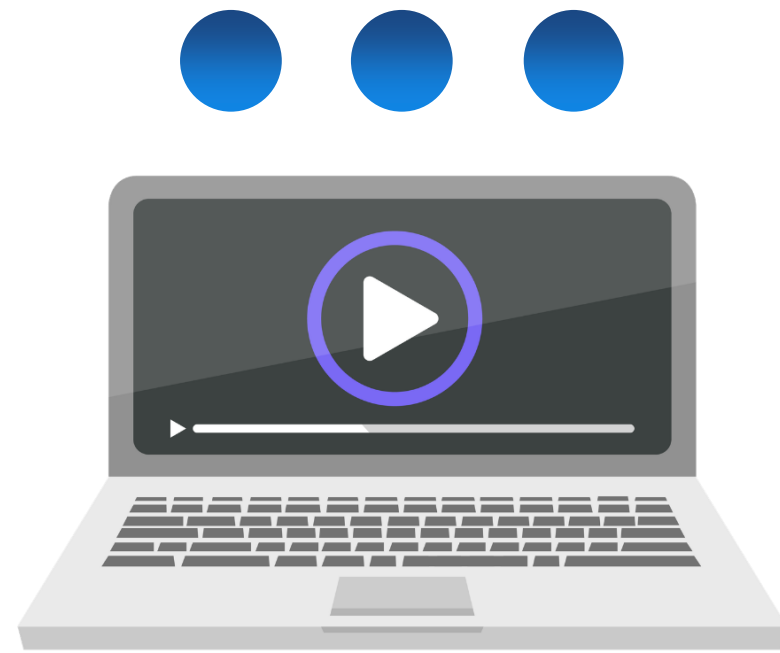
Demo

Workspace Overview



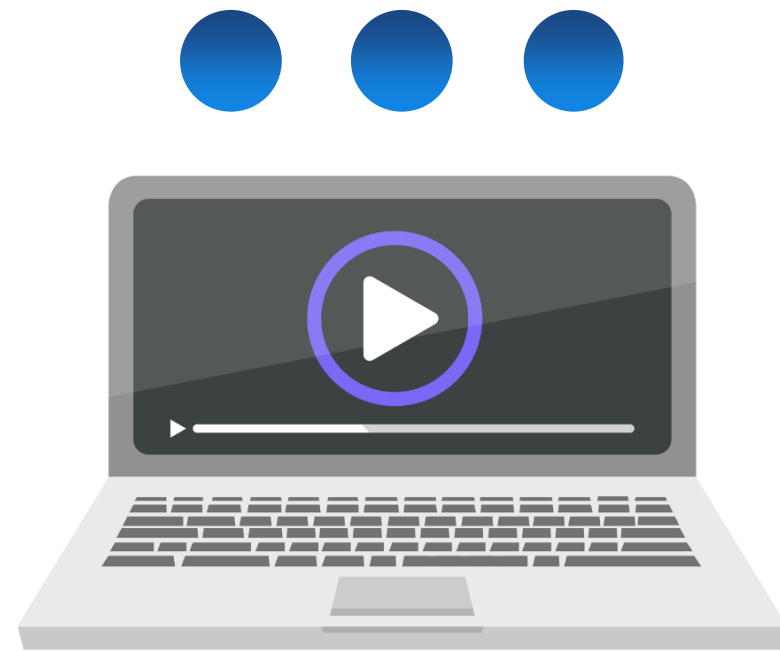
Demo

Adding AWS Keys as Environment Variables



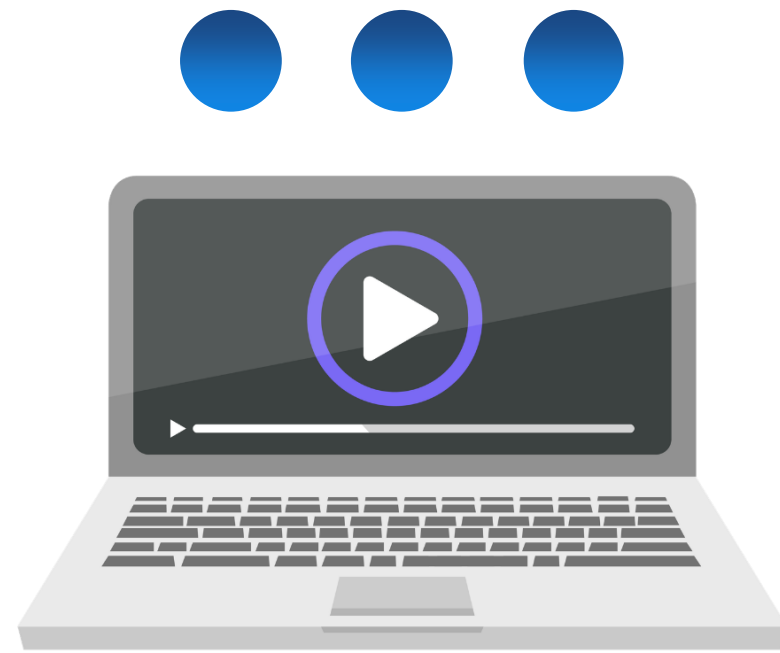
Demo

Running Plan Manually



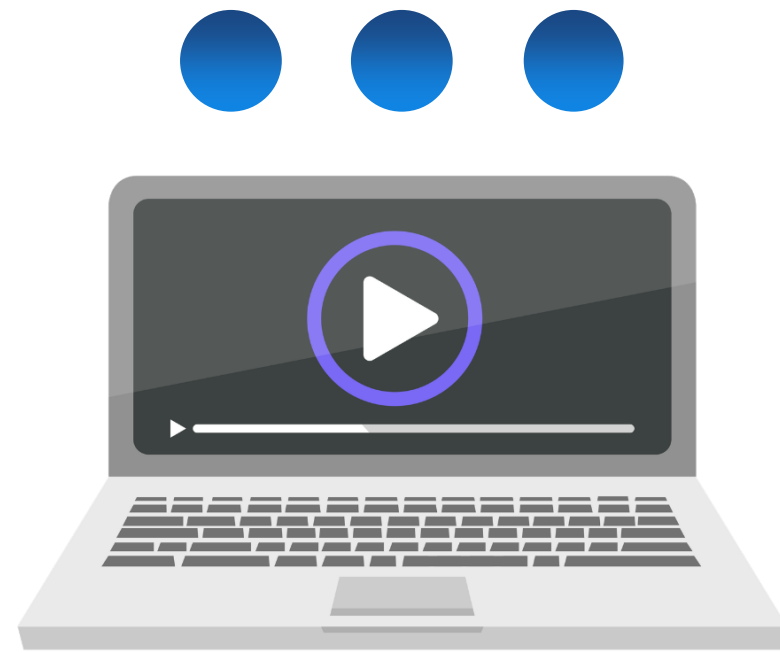
Demo

Adding Terraform File



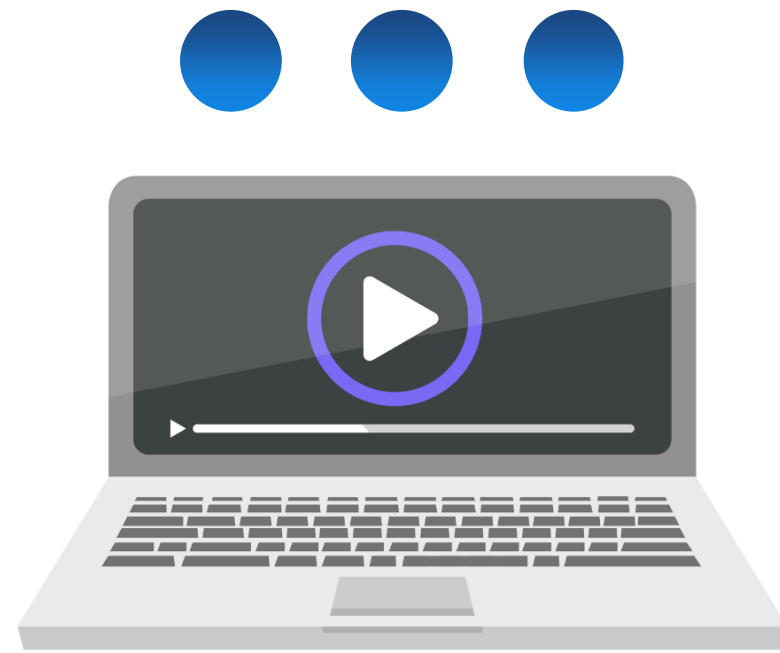
Demo

Auto Apply the Changes



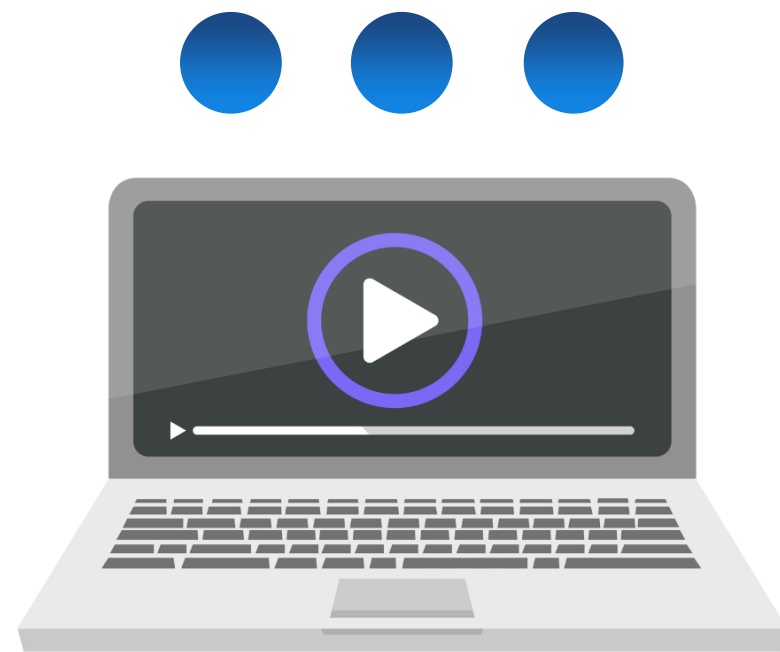
Demo

Automatic speculative plans on PR



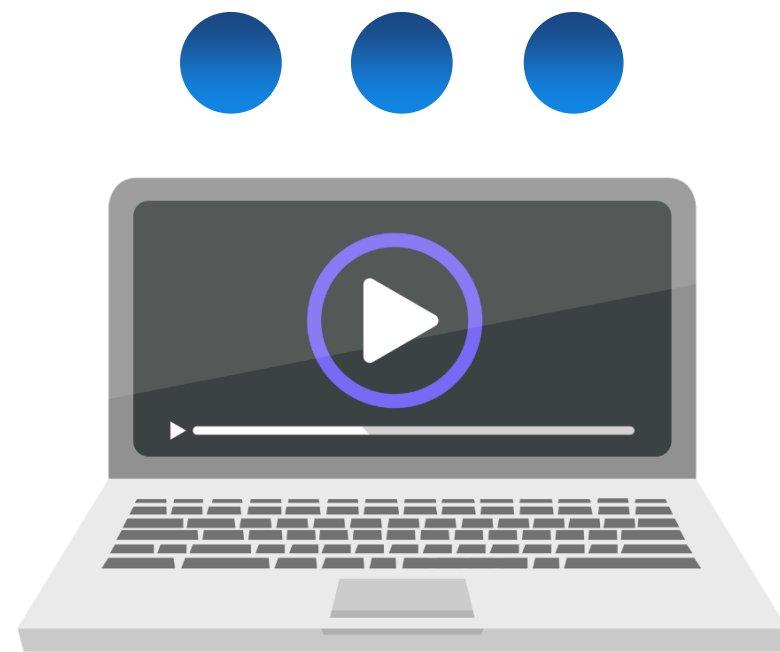
Demo

Enabling Cost Estimation



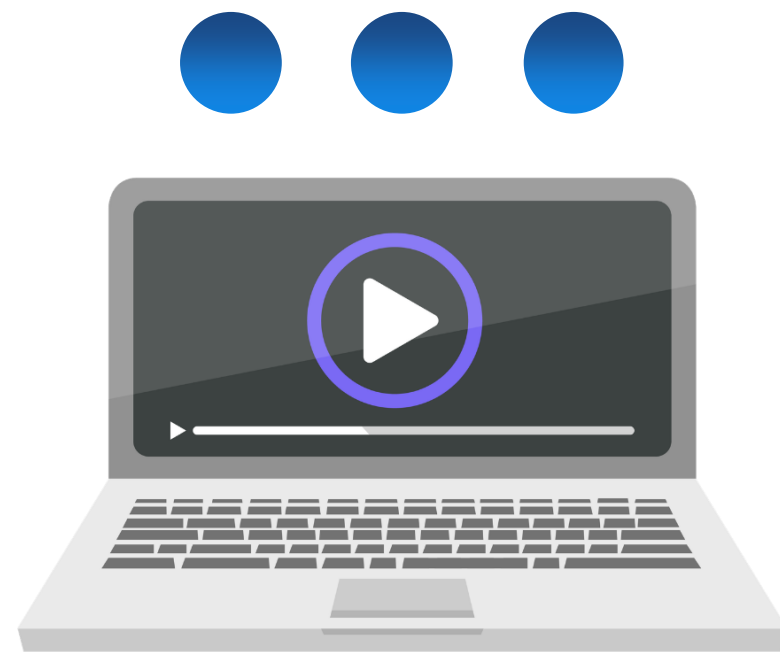
Demo

Automatic Speculative Plans with Cost Estimations



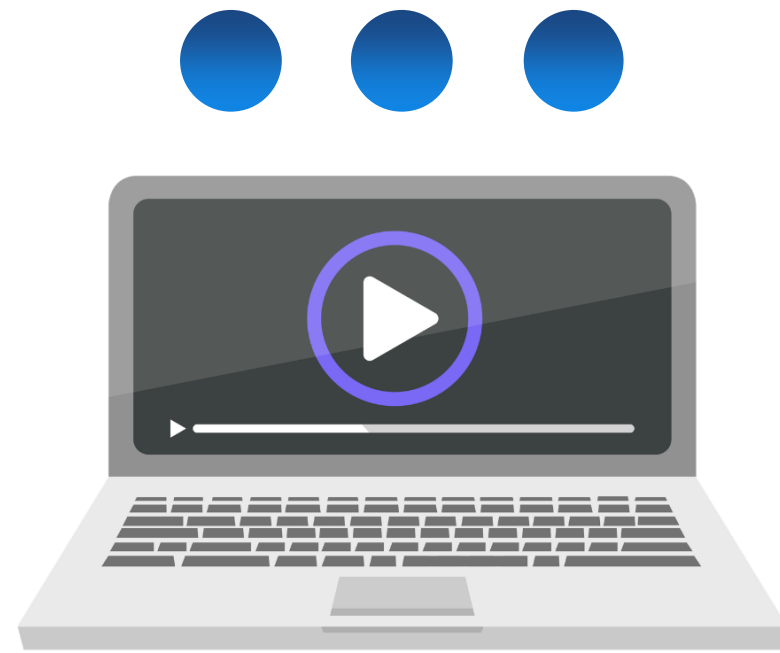
Demo

Adding Terraform Variables Normal Format



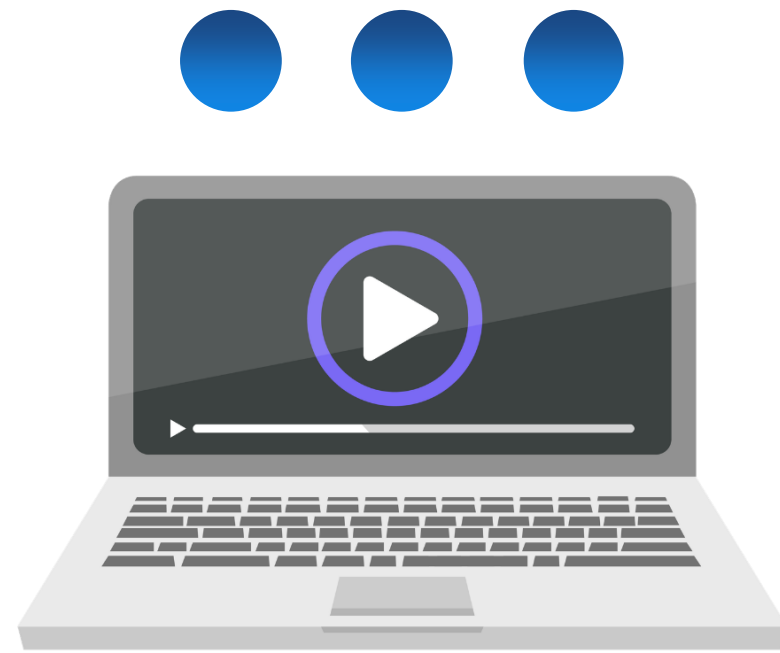
Demo

Adding Terraform Variables with HCL Format



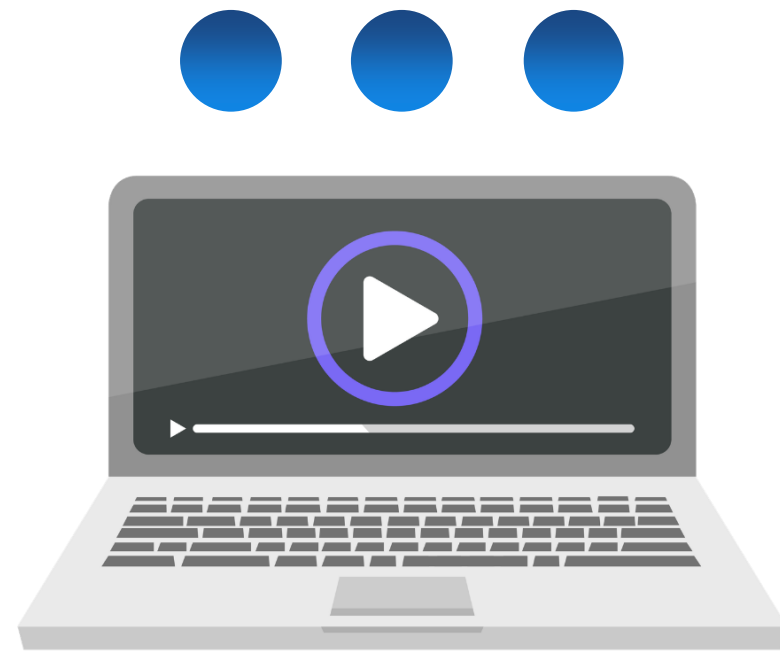
Demo

Sending Notifications to MS Teams - 1



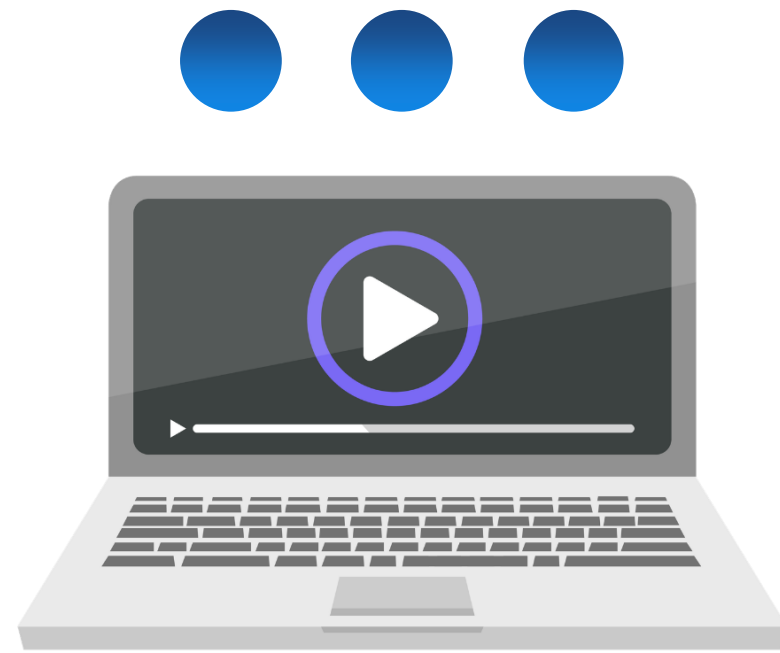
Demo

Sending Notifications to MS Teams - 2



Demo

Access Management



Demo

Resource & Workspace Deletion



THANK YOU

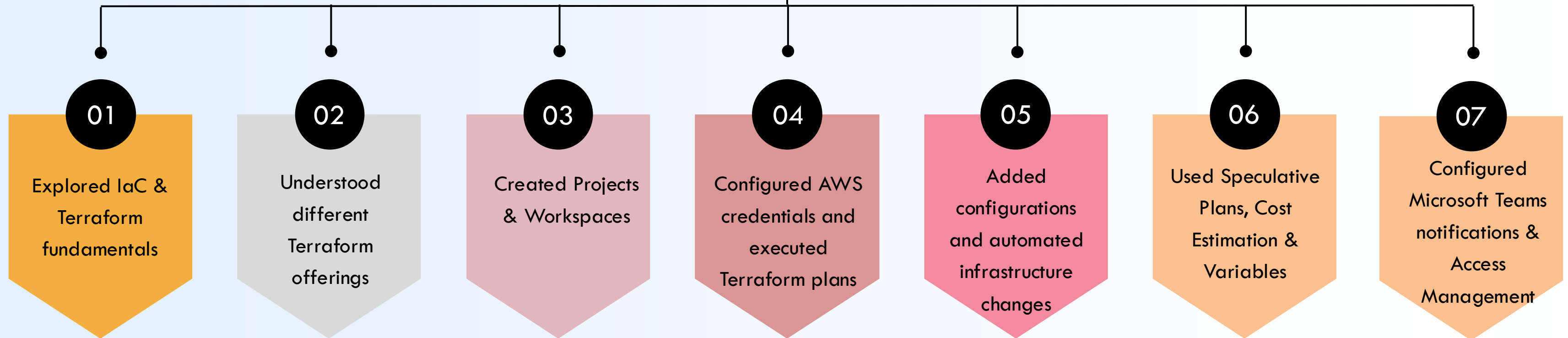
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Conclusion

Mastering Terraform Cloud - A Hands-on Beginners Guide





Built a foundation for centralized Terraform workflows

Next: Remote State, Teams & Permissions, VCS Workflows, Policy as Code, Private Registries & CI/CD



8n AI Agents Bootcamp
Cloud & SRE Projects



KIND 101: Learn Kubernetes in Docker -
No Edits, Just Real Demos

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Plan for the Absolute Beginners -
s-On DevOps



Mastering Docker Essentials - Hands-On
DevOps



Master Linux Bootcamp for
SRE & Cloud Engineers



Infrastructure Automation with
OpenTofu - Hands-On DevOps

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Practical Kubernetes - Beyond CNA and
CKAD | Hands-on



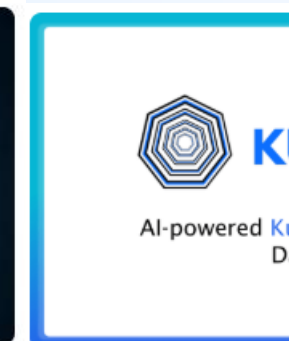
Argo CD for the Absolute Beginners -
Hands-On DevOps

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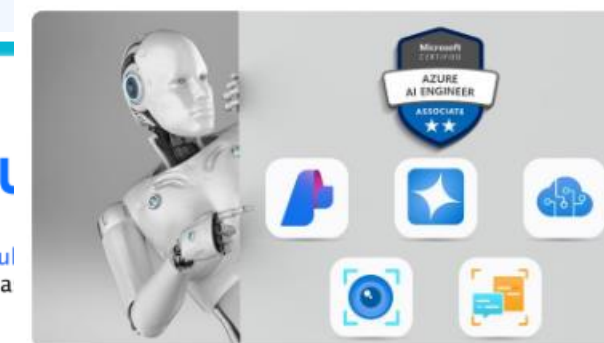


Hands-On with AWS Cloud - A
Beginners Guide

es



KubeBuddy - AI-powered
Kubernetes Monitoring Dashboard



Microsoft Certified Azure AI Engineer
Associate : AI-102

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DevOps Bootcamp for All

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Git & GitHub Bootcamp - Build, Track &
Collaborate | Hands-On

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Mastering Amazon EKS

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Hands-On CI/CD with GitHub Actions |
Absolute Practical



Getting Started with AI



Kubernetes and Cloud Native Associate
(KCNA) + Practice Exam

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Cloud Native for
Beginners

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Getting Started with Machine Learning
using Python | HandsOn

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Data Science - From Data to



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